

by varying $p \in [1, \infty)$ one gets (many?) non-trivial invariants of Γ .

8.5.Y Semistability of the action of Γ on $\partial\Gamma$: Let a word hyperbolic group Γ with a fixed generating set $G \subset \Gamma$ act on a metric space D . Fix $\epsilon > 0$ and $\lambda > 1$ and call $\gamma \in \Gamma$ *expanding* at a given point $p \in D$ if γ^{-1} sends the ball $B(\gamma p, \lambda\epsilon) \subset D$ into the ball $B(p, \epsilon) \subset D$. Then define the subset $\psi_p \subset \Gamma \times \Gamma$ of the pairs (γ_1, γ_2) such that $\gamma = \gamma_1 \gamma_2^{-1}$ is expanding and contained in G . Call the action of Γ on D *regular expanding* if this ψ_p is a quasigeodesic nonexpanding field on Γ in the sense of 7.6 for some G , ϵ and λ and all $p \in D$. (Strictly speaking, the "quasigeodesic non-expanding" applies to the Cayley graph X of (Γ, G) .) Notice that the action of Γ on $\partial\Gamma$ is regular expanding for the metric on $\partial\Gamma$ constructed in 7.2. Furthermore, small perturbations of a regular expanding action on a compact space D (obviously) are regular expanding.

Now, according to 7.6, every quasigeodesic non-expanding field on Γ defines a unique point in $\partial\Gamma$. Hence, there is a continuous Γ -equivariant map $D \rightarrow \partial\Gamma$ for every regularly expanding action of Γ on D . In particular *the action of Γ on $\partial\Gamma$ is semistable*. Namely, for every small perturbation of this action, say $(\partial\Gamma, \Gamma)_{\text{pert}}$, there exists a continuous Γ -equivariant map $S: (\partial\Gamma, \Gamma)_{\text{pert}} \rightarrow (\partial\Gamma, \Gamma)$. Clearly such an S is unique and converges to the identity as the perturbation goes to zero. (This semistability result in a geometric context is due to Sullivan, see [Sul].)

The case of groups Γ of rank ≥ 2 . Let X be symmetric space of non-positive curvature, let ∂X be the space of asymptotic classes of Weil chambers and let Γ be a discrete cocompact isometry group acting on X and thus on ∂X . By moving (compact pieces of) chambers from X to a fixed Γ -orbit in X one can easily produce "quasiflat non-expanding fields" in Γ and show that the action of Γ on ∂X admits a Markov presentation (over Γ) and that this action is semistable. However, one does not know how to make such a construction quasi-isometry invariant if $\text{rank}_{\mathbb{R}} X \geq 2$. One also does not know how to find a Markov presentation of ∂X which is finite-to-one and how to count words (and conjugacy classes

of Abelian subgroups) in Γ .

8.6 Relative hyperbolicity: Let us give a hyperbolic version of small cancellation theory over free products by adopting geometric language of manifolds with cusps.

Let X be a complete hyperbolic locally compact geodesic space with a discrete isometric action of a group Γ such that the quotient space $V = X/\Gamma$ is quasi-isometric to the union of k copies of $[0, \infty)$ joined at zero. To simplify the matter we assume the action of Γ on X is free and then lift the k rays in X (corresponding to the k points in $\partial V \approx \{1, 2, \dots, k\}$) to k rays $r_i: [0, \infty) \rightarrow X$, $i = 1, \dots, k$. Denote by h_i the corresponding (ray) horofunctions and by $r_{i(\infty)} \in \partial X$ the limit points of r_i . Denote by $\Gamma_i \subset \Gamma$ the isotropy subgroups of $r_{i(\infty)}$ for the action of Γ on ∂X and assume that Γ_i preserves h_i for $i = 1, \dots, k$. Denote by $B_i(\rho)$ the horoballs $h^{-1}(-\infty, \rho) \subset X$ and assume that for a sufficiently small ρ the intersection $\gamma B_i(\rho) \cap B_j(\rho)$ is empty unless $i = j$ and $\gamma \in \Gamma_i$. Denote by $\Gamma B(\rho) \subset X$ the union $\bigcup_{i, \gamma} \gamma B_i(\rho)$ over $i = 1, \dots, k$ and all $\gamma \in \Gamma$, set $X(\rho) = X \setminus \Gamma B(\rho)$ and assume the action of Γ on $X(\rho)$ is cocompact for all $\rho \in (-\infty, +\infty)$.

8.6.A Definition: A group Γ is called word hyperbolic relative to some subgroups $\Gamma_1, \dots, \Gamma_k$ in Γ if Γ admits an action on some X with the above properties, where Γ_i are the isotropy subgroups of h_i .

Examples:

- (a) Take the free product of some groups Γ_i and add finitely many relations which are "generic sufficiently long" words in $\gamma_i \in \Gamma_i$. The resulting group Γ is hyperbolic relative to Γ_i .
- (b) Take a group Γ with a subgroup $\Gamma_0 \subset \Gamma$ and let $\gamma_1, \dots, \gamma_m$ in Γ_0 generate Γ_0 and $\gamma_1, \dots, \gamma_m, \dots, \gamma_{m+n}$ generate Γ . Let $|\cdot|$ be the maximal left invariant metric in Γ for which $|\gamma_i| \leq 1$

for $i = 1, \dots, m+n$ and $| \gamma | \leq \log(1 + | \gamma |_0)$ for all $\gamma \in \Gamma_0$, where $| \cdot |_0$ is the word metric in Γ_0 . If $(\Gamma, | \cdot |)$ is hyperbolic and the embedding $(\Gamma_0, \log(1 + | \cdot |_0)) \rightarrow (\Gamma, | \cdot |)$ is quasi-isometric then Γ is word hyperbolic relative to Γ_0 .

(c) Let Γ be a finite covolume discrete isometry group of a complete simply connected Riemannian manifold X with *pinched* negative curvature,

$$0 > -a \geq K(X) \geq -b > -\infty.$$

Then the quotient space $V = X/\Gamma$ is quasi-isometric to the wedge of several copies of $[0, \infty)$ and Γ is hyperbolic relative to the (cuspidal) isotropy subgroups of $r_i(\infty) \in \partial X$. For example, every finite covolume lattice in a simple Lie group with $\mathbb{R}$ -rank = 1 is hyperbolic relative to the cuspidal subgroups.

(d) The relative hyperbolization in 3.4.C gives us a group $\pi_1(H)$ which is hyperbolic relative to $\pi_1(P_0)$.

8.6.B All our results on word hyperbolic groups extend with minor modifications to the relative case. Here we only look on the relative version of Markov coding of Γ .

8.6.B₁ Definition: A graph Y is called *geo-Markov* if for every finite subset of vertices $Y_0 \subset Y$ there exists a finite union $\tilde{Y}$ of Markov trees of finite types and a simplicial map $\cap: \tilde{Y} \rightarrow Y$ bijective on the set of vertices, such that every point $\tilde{y} \in \tilde{Y}$ can be joined by a path (in $\tilde{Y}$) with some vertex $\tilde{y}_0 \in \tilde{Y}_0 \stackrel{\text{def}}{=} \cap^{-1}(Y_0)$ and

$$\text{dist}(\tilde{y}, \tilde{Y}_0) = \text{dist}(y, Y_0)$$

for all $\tilde{y} \in \tilde{Y}$ and $y = \cap(\tilde{y})$. Next, a group Γ is called *geo-Markov* if every locally finite graph Y with the edges of integral length which admits a cocompact isometric action of Γ is

geo-Markov. Clearly, every geo-Markov group Γ has a *rational* word counting Z-function.

8.6.C Examples:

(a) We know already that every word hyperbolic group is geo-Markov.

(b) It is likely that *every Abelian group is geo-Markov*. To see this we order the edges issuing from $x \in Y$, for all vertices $x \in Y$, such that this local order is Γ -invariant. This gives us as earlier, a (lexicographic) order on shortest paths from x to Y_0 for all vertices x in Y . The first path for this ordering consists of at most N "pieces" for some N depending on (Y, Γ) and Y_0 , where every piece P_i , $i = 1, \dots, N$ is the composition of Γ -translates of a fixed path P_1 .

$$P_i = \gamma_1 P_1 * \gamma_2 P_1 * \dots * \gamma_m P_1,$$

for some m depending on the original shortest path. Therefore, this path looks like a broken line with at most N break-points, where the "segment" between two breaks is parallel to one of N given "directions" p_i . The "type" of such a path is given by a finite sequence of directions. Now every $x \in Y$ is characterized by *finitely many* possible types of first shortest paths from z to Y_0 via x for all $z \in Y$ and the geo-Markov property becomes feasible.

8.6.D Proposition: *Let Γ be word hyperbolic relative to some subgroups Γ_i , $i = 1, \dots, k$, in Γ . If all Γ_i are geo-Markov then Γ also is geo-Markov. In particular, the counting function for $\#\{\gamma \in \Gamma \mid |\gamma| = k\}$ is rational for every word metric in Γ . (This suggests an approach to the rationality question in [Can] for lattices in $O(n, 1)$.)*

Proof: Let us look at the geometry of $X(\rho) \subset X$ defined at the

beginning of this section. Take a segment $[x_1, x_2]$ in X and let $[x_1, x_2]_\rho$ be the union of $[x_1, x_2]$ with the Γ -translates of the horoballs $B_1(\rho)$ which intersect $[x_1, x_2]$. Then one easily sees with the convexity analysis in 7.3 that the subset $[x_1, x_2]_\rho \subset X$ is ϵ -convex, where ϵ only depends on X and ρ . Therefore, if x_1 and x_2 lie in $X(\rho)$ then every shortest path in $X(\rho)$ between x_1 and x_2 (if such path exists) lies ϵ' -close to $[x_1, x_2]_\rho$ where ϵ' depends only on X and ρ . Moreover, the same holds true for quasigeodesics in $X(\rho)$ for the geodesic metric defined by the lengths of minimal paths in $X(\rho)$ between pairs of points. Now, we may assume $X(\rho)$ equals our graph Y as the two are quasi-isometric. Then two shortest paths issuing from nearby points in Y and heading for Y_0 stay close unless they follow for some time same Γ_i -orbit or its Γ -translate. Then, as the paths go away from such an orbit, they again become mutually close. Therefore, the geometry of $C(Y_0, x)$ is determined by the pattern of Γ -translates of Γ_i -orbits lying "behind" x (as looked upon from Y_0) and by the geometry of paths in Y_0 . The Markov property for the former follows by our "hyperbolic" discussion in 7.6 and the latter is taken care of by "geo-Markov" for Γ_i . Making this geometric discussion rigorous is left to the reader.

Questions: Are finitely generated nilpotent groups geo-Markov? Are Cartesian products of geo-Markov (i.e. word hyperbolic) groups geo-Markov? Are 1/5-groups (and semihyperbolic groups, in general) geo-Markov? What are examples (if there are any) of Markov groups (see 5.2.E) which are not geo-Markov?

Remark: Jim Cannon pointed out to me that the stronger Γ -invariant (see 8.5.) version of the geo-Markov property may fail for Cayley groups of finite extension of Abelian groups (and also of nilpotent groups). His criticism forced me to modify the definition and to leave the above (b) inconclusive.

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